# MANISH KUMAR

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## PROFESSIONAL SUMMARY

Supply chain and analytics person with a STEM MBA and about seven years running an $18M operations portfolio before business school. I spent those years getting messy operational data into shape and using it to make calls on planning, sourcing, inventory, and delivery, and I did a live consulting project at Babson on top of that. I like the work where you frame a fuzzy problem, build the model or the fact base, and hand a leader something they can actually decide on. I move between strategy, analytics, business analysis, and project controls without much friction, I can walk a senior audience through both the numbers and the story, and I build with current AI tools like IBM watsonx and Microsoft Copilot.

## EDUCATION

**Babson College, F.W. Olin Graduate School of Business | Wellesley, MA** — **Sep 2024 to May 2026**
*STEM MBA, Business Analytics and Machine Learning, Supply Chain Analytics*

**Manav Rachna International Institute of Research and Studies | Faridabad, India** — **Jul 2013 to Aug 2017**
*B.Tech, Civil Engineering*

## CORE SKILLS

**Supply Chain and Operations:** Plan, source, make, deliver, and return; demand and supply planning and S&OP; material requirements planning; inventory optimization, EOQ, safety stock, and reorder points; replenishment and joint replenishment; capacity planning; network design and optimization with MILP; facility location and total landed cost; transportation and route analysis; supplier performance, sourcing, and vendor management; procurement, RFQs, and quotation analysis; inbound and outbound logistics; supply resiliency, risk, and contingency planning; exception and shortage management

**Strategy and Consulting:** Problem framing and fact base work; market and competitive analysis; TAM, SAM, and SOM sizing; customer segmentation and personas; willingness to pay and price sensitivity; go to market strategy; business cases and scenario modeling; benchmarking; executive communication; stakeholder influence

**Data, Analytics, and Optimization:** Python with pandas, NumPy, scikit-learn, PuLP, and Gurobi; R; SQL with joins, CTEs, window functions, and subqueries; optimization and MILP; ARIMA forecasting and time series; statistical and hypothesis testing; Monte Carlo simulation; regression, sensitivity, and scenario analysis; root cause and variance analysis; classification, feature engineering, and model evaluation

**AI, GenAI, and NLP:** IBM watsonx.ai, watsonx Orchestrate, and Granite and Slate models; Microsoft Copilot Studio; Model Context Protocol; agentic workflows, tool calling, and structured outputs; prompt and context design; REST API integration; embeddings, cosine similarity, TF-IDF, UMAP, and semantic classification; RAG concepts; retrieval and model evaluation

**Business Intelligence and Reporting:** Power BI, Tableau, DAX, Power Query, and Power Pivot; advanced Excel with PivotTables, SUMIFS, XLOOKUP and VLOOKUP, nested formulas, conditional formatting, and data validation; KPI dashboards; executive and automated reporting; trend, drill down, and data storytelling

**Data Engineering and Governance:** ETL and ELT; Apache Airflow; Databricks; Google BigQuery; SQLite and DBeaver; relational data modeling, ERDs, and third normal form; data validation, reconciliation, and duplicate detection; master data governance; data dictionaries, business rules, and exception thresholds; referential integrity, auditability, and UAT

**Enterprise and Project Systems:** SAP MM, PS, FI/CO, and PM, with SAP IBP and Kinaxis RapidResponse familiarity; ERPNext, Farvision ERP, and Tally; TMS; Primavera P6 and Microsoft Project; Jira and Confluence; Git and GitHub; Agile and Scrum, Waterfall, and hybrid; PMO and portfolio governance, WBS, and risk and change control

**Project and Process:** Business process analysis; requirements gathering and documentation; process mapping and standardization; fit gap analysis; UAT and test scripts; continuous improvement; cross functional coordination; executive business reviews

## PROFESSIONAL EXPERIENCE

**NS Constructions Pvt. Ltd.- Project Manager | Supply Chain, Data and Business Analyst** — **Apr 2021 to Jul 2024**
*Patna, India. $18M capital portfolio, 10+ concurrent infrastructure projects, 55-person cross functional team.*

- Ran material planning, procurement, inventory, logistics, equipment, and project controls across an $18M portfolio of more than ten projects at once, coordinating the whole thing through a 55-person team that cut across engineering, procurement, finance, contractors, and the field.
- Pulled SAP MM, PS, and FI/CO together with the ERP, the TMS, and Primavera P6 so material requirements, purchase orders, goods receipts, inventory, supplier commitments, transportation status, cost, and schedule all sat in one place instead of a dozen scattered files.

- Built the material and replenishment plans off project schedules, past consumption, inventory on hand, and supplier lead times, then checked them against open orders and site schedules so shortages and timing risks showed up before they hit the field.
- Made the Excel and Power BI dashboards leadership actually used, tracking inventory health, supplier performance, committed cost, budget variance, and schedule. That work cut reporting turnaround by 40% and moved on time reporting from 75% to 95%.
- Set the master data definitions, KPI logic, validation rules, and reconciliation steps across all ten-plus projects, which is dull work but it is what made the numbers consistent enough to trust.
- Chased down the recurring problems, supplier delays, inventory that did not match, odd consumption, transportation snags, and cost variances, and the fixes that came out of it cut repeat exceptions by about 24%. One of those calls kept a 14-day work stoppage from happening during COVID.
- Wrote the contingency plans for when things went wrong, monsoon flooding, late suppliers, transportation shutdowns, shortages, equipment failures, using backup sourcing, load consolidation, and schedule resequencing to keep critical path work moving.
- Leaned on Primavera P6 work packages and milestone reviews to catch schedule and capacity problems earlier, which lifted milestone adherence about 12% and saved close to 30 days of delay. Reworking the procurement to delivery flow trimmed PO processing about 20% and issue resolution about 28%.
- Gathered requirements for a digital measurement and billing platform on Farvision ERP, turned manual field and finance workflows into data fields and reporting rules, and ran the UAT that brought implementation defects down 25%.
- Worked shoulder to shoulder with procurement, finance, engineering, vendors, and leadership, including World Bank and government stakeholders, on reviews and budgeting, and rolled out a QA and QC system that helped cut delays 15% and save roughly $500K a year.

**ChopValue Boston - Student Consultant (Babson Consulting Experience)** **Jan 2025 to May 2025**

*Boston, MA. Live strategy and analytics engagement for a certified B Corp making carbon negative recycled panels.*

- Started with a hard question, how do you grow demand for a niche premium product when there is barely any public data, and broke it into pieces covering customers, pricing, market size, and competitive positioning. The research approach was mine to build from scratch.
- Ran 20+ in depth interviews with architects, designers, and institutional buyers, more than 20 questions each, which gave me 130+ responses on what people will pay and how they actually pick a vendor. Built the guides in Qualtrics and Google Forms.
- Benchmarked more than 50 competitors on price, performance, sustainability, lead time, and delivery, and scored 17-plus alternatives on a five parameter rubric with a price versus sustainability map.
- Sized the market top down with IBISWorld and Statista (TAM $28.8B, SAM $2.5 to 4B, SOM $20 to 30M) and built a pricing and elasticity model that turned up a 40% willingness to pay premium, which changed how the client thought about pricing.
- Turned all of it into a three tier segmentation, a $60 to 70 per square foot pricing recommendation with a phased increase tied to conversion, and a 12-month go to market plan, then took it straight to the owner and faculty advisors.

**Whale Blue Hitech Pvt. Ltd. - Assistant Engineer** **Oct 2017 to Mar 2021**

*Patna, India. Multi site operations, reliability, and reporting.*

- Led the development of watershed modeling projects to improve hydrological efficiency and water quality for over 10,000 residents.
- Mentored 21 junior engineers, conducting on-site workshops to enhance technical capabilities and boost project efficiency. Supervised inspection and maintenance of flood control structures, increasing operational uptime and extending embankment life.
- Conducted terrain analysis and hydrological assessments using SLIDE, GEO SLOPE, and HEC-RAS for data-driven infrastructure design.
- Worked on operational improvements across several sites, digging through downtime and failure data to standardize how the work got done, which brought asset failure rates down 22%.

## SELECTED PROJECTS

***Supply Chain, Network Design, and Optimization***

- **Distribution Network and Total Cost Optimization (EcoGoods)** *(Python, Gurobi, PuLP, pandas, NetworkX, Folium, geopy).* Built a five year network model for 50,000 pallets across six markets, six candidate distribution centers, and 42 lanes, working out total landed cost from real freight rates and Haversine distances under demand and capacity limits. I compared single site, dual site, and a few other setups and landed on a $3.24M Baltimore hub, which came in 29.6% under the best single facility option and 42.7% under a dual facility network. I also worked out the demand point where a second site finally starts to pay off.

- **Reverse Logistics Network and Emissions Optimization** *(Python, Gurobi, PuLP, NetworkX, Folium).* Modeled a return network across 11 retail locations, 10 to 19 consolidation points, and several partners on live distance data, with vehicle capacity, routing distance, sorting cost, and a carbon limit built in. Laid the cost versus service and cost versus emissions trade offs out on a Pareto frontier and delivered a routing plan you can re run as things shift. It cut transportation miles about 20%, operating cost about 12%, and emissions about 28%, and pulled vehicle utilization well up from a weak starting point.
- **Supplier Consolidation and Cost Reduction (TechWave)** *(Python, EOQ, joint replenishment).* Took a $2.2M component portfolio that had scattered ordering, high transaction cost, and too much single supplier reliance, and modeled ordering, holding, and purchasing cost across item level EOQ and joint replenishment. The interesting result was that full consolidation would cost more in holding than it saved in ordering, so I recommended a hybrid: consolidate the predictable high volume parts, keep the specialists for the critical ones.
- **Inventory and Supplier Lead Time Optimization (ColorWeave)** *(Python, EOQ, safety stock, reorder points).* Modeled order quantities, reorder points, and safety stock across five critical SKUs from two suppliers, weighing a responsive five day supplier against a cheaper 20 day one and showing how much the longer lead time drives up safety stock and holding cost. Balancing purchase cost, lead time, and stockout risk saved about $920 while roughly doubling the buffer on the materials most likely to cause trouble.
- **Supply Planning and Exception Management Pipeline** *(Apache Airflow, SQL, Python, pandas).* Built a planning workflow that pulls inventory, demand, open POs, and supplier lead times together, validates and reconciles the data, works out days of supply, and ranks shortage risk. Airflow reruns it on a schedule, so a big pile of supply records becomes a short daily list of what actually needs attention.

***AI, GenAI, and Semantic Analytics***

- **Primo AI Library Assistant, Two Platform Build and Evaluation** *(IBM watsonx Orchestrate, Microsoft Copilot Studio, IBM Granite, MCP, Node.js and Express, IBM Cloud Code Engine, REST API).* Built and shipped a live AI library assistant on two competing platforms on my own, wiring a custom Node.js and Express MCP server to Babson's Ex Libris Primo API with server sent events, API key auth, and secure config. Set up three auto discovered tools for search, resource type filtering, and record metadata, all grounded in real catalog data so it would not make things up. Did the stakeholder discovery, built an evaluation harness from 1,564 real queries collected over eight months (26 of 29 test cases passed, about 90%), and wrote the rollout plan. It was mine end to end, from the architecture through the evaluation.
- **Semantic Market Intelligence and Narrative Drift Analysis** *(Python, IBM watsonx Slate, embeddings, cosine similarity, UMAP, TF-IDF).* Went through 55,108 documents from 2011 to 2025 to see how one public figure's messaging drifted over time, using 768 dimensional Slate embeddings and anchor based cosine similarity instead of topic modeling. UMAP made the clusters readable, TF-IDF was the second opinion, and t-tests (p = 0.00) confirmed the shift was real rather than just more volume. One theme climbed from around 40% to 94% of the emphasis.
- **Multi Model AI Evaluation Benchmark** *(IBM Slate and comparison embedding models, Precision@5, silhouette, throughput).* Benchmarked five embedding models over 55,108 documents on retrieval quality, clustering, and speed to figure out which one is worth deploying. The useful surprise: the model with the worst clustering still hit perfect Precision@5, which is a good reminder that a headline metric can fool you if you do not tie it back to the real decision.

***Data Engineering, ML, and Forecasting***

- **Credit Risk and Default Prediction Model** *(Python, scikit-learn, R).* Built a classifier on 150K imbalanced consumer loan records to predict default, using 1:14 class weighting and comparing logistic regression, decision tree, random forest, a neural net, and an ensemble. I picked the model on the business cost of a missed default instead of raw accuracy, getting to 77.8% sensitivity and 0.86 AUC, because here a false negative hurts far more than a false positive.
- **Python and SQL Automation Pipeline** *(Python, pandas, SQLite, SQL, REST API).* Built a pipeline that pulls from a REST API and flat files into a constraint enforced SQLite schema, replacing a manual reporting slog. Added validation, duplicate detection, and integrity checks, ran the same aggregations in both SQL and plain Python to cross check, and automated the reports. It surfaces revenue, orders, and margin ($4,651 revenue, $46.51 average order value, 52.87% margin) and just reruns cleanly.
- **Relational Database and BI Analysis** *(SQL, SQLite, DBeaver, ERD, third normal form).* Designed a nine table schema in third normal form with a full ERD covering projects, people, tasks, expenses, and budgets, and loaded 572 records with zero integrity violations. Wrote layered SQL with joins, CTEs, and window functions to line up $14.57M in budgets against $1.29M in actual spend, with orphan key, null, and range checks keeping the analytical layer clean.
- **Economic and Operational Forecasting** *(R, ARIMA, time series analysis).* Built forecasts across five-plus economic and operational series, testing stationarity with ADF and KPSS, reading structure off ACF and PACF, and comparing ARIMA specs with AIC and BIC. Ljung-Box diagnostics and 80% and 95% prediction intervals helped me separate trends worth acting on from forecasts too shaky to trust.

***Finance, Strategy, and Commercial Analytics***

- **Corporate Valuation and Investment Decision Model (Newmont)** *(Excel, DCF, CAPM, Monte Carlo, @RISK, regression).* Built a risk adjusted valuation to decide whether shareholders should take a $75 per share offer, combining DCF, CAPM, beta estimation, regression, and a 50,000 iteration Monte Carlo run. It came out to a mean of $78.02 with a $33.75 standard deviation and a 54% chance of ending up below the offer, so I said take the deal. The small upside was not worth that much downside.
- **Strategic Financing and Investment Decision Analysis** *(Excel, PrecisionTree, decision trees).* Built a decision model for a $300K expansion under three financing options, organic, bank loan, and equity partnership, using decision trees, tornado charts, and sensitivity analysis to compare expected values of $1.34M, $920K, and $424K. The equity partnership won on both expected value and risk.
- **Macroeconomic Analysis, Japan Reshoring** *(economic modeling, IMF and OECD data).* Worked through six macro indicators, GDP, productivity, inflation, labor, currency, and trade, to see how the bigger economic picture pushes sourcing and manufacturing decisions, and tied that back to real supply chain choices.
- **Wayfair Customer Segmentation and Campaign Analytics** *(Tableau, Excel, A/B testing, EDA).* Checked whether Wayfair's Ship in Time holiday guarantee actually moved sales, breaking down segments, order frequency, platforms, categories, and delivery outcomes. Activated Customers were 57.09% of the base, and the guarantee did little overall but landed better with new and returning visitors, which pointed toward more targeted incentives.
- **E-Commerce Logistics, Freight, and Seller Performance Dashboard (Olist)** *(Tableau).* Built an interactive dashboard on a nine table model over roughly 370,000 records joined on eight keys, with calculated fields, geographic heatmaps, seller scorecards, and freight versus order value views, plus cross dashboard actions so anyone could pull up the high cost lanes, the delayed categories, and the weak service spots.
- **Streaming Industry Strategy** *(Value Net and PARTS framework).* Led a strategy workstream sizing up the competitive landscape and the growth vectors, then pulled it into a tight memo for executives. A clean example of structured qualitative analysis under a real framework.

## CERTIFICATIONS

Introduction to Management Consulting (Emory) · Data Analytics (Google) · Academy Accreditation, Generative AI (Databricks) · Complete SQL Bootcamp · PwC US Management Consulting Job Simulation